

A Radiology

Input Data



Radiography
2D projection



Ultrasound
real-time video



MRI
3D · T1/T2/DWI

Dimensionality
2D projection · quasi-3D (DBT) ·
3D volumetric (CT, MRI, PET) ·
2D + time (ultrasound, cine MR)
· 4D (DCE-MR, 4D CT)

Paired non-image data
report text · DICOM metadata ·
clinical and staging · molecular



Mammography
2D · DBT



CT
3D · structural



PET
molecular · SUV

Preprocessing

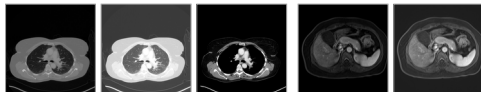


DICOM parse,
series selection,
de-identify



scanner / site
harmonization
(ComBat)

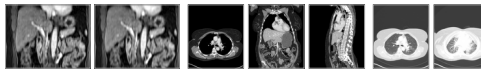
Intensity standardization



CT: window / level
full range, lung, mediastinum

MRI: z-score, N4 · PET: SUV

Geometry

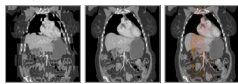


resampling
3.0 mm to 1 mm isotropic

orientation and 2.5D
3 planes, adjacent slices

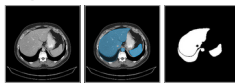
MIP (optional)
slice to 60 mm slab

Registration



longitudinal · inter-modality · atlas
rigid / deformable

Segmentation to ROI / VOI



anatomy crop, then lesion VOI
also a clinical task

Models

① Pre-defined features



real VOI · its HU histogram · its 32-bin GLCM

IBSI: first-order, shape,
texture (GLCM, GLRLM,
GLSZM, NGTDM),
wavelet / LoG

selection +
classical ML
(LASSO, RF, Cox)

② Supervised deep learning



CNN



U-Net
encoder-decoder



ViT

③ Self-supervised and foundation



SSL
pretraining



foundation
model



vision-
language

Task head



linear · Cox · segmentation decoder ·
text decoder

one backbone, many tasks; the head decides which

Clinical Tasks

along the care pathway



Screening /
future risk



Triage /
workflow



Detection /
localization



Segmentation /
quantification



Diagnosis /
characterization



Radiogenomics /
molecular



Treatment
planning



Response /
longitudinal



Prognosis /
survival



Report
generation / VQA

the same segmentation, once as ROI extraction and once as an endpoint